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SKILLING-ML/DL

II-II CSE, CSE(AI&ML)

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Decision Tree (ID3 Algorithm) – Solved Example

What is a Decision Tree?

A Decision Tree is a supervised learning algorithm used for:

- Classification
- Regression

It splits data based on feature values to create a tree-like structure of decisions.

Invented and popularized for machine learning by researchers like
Ross Quinlan (ID3, C4.5)

Core Concept Behind Decision Trees

Tree building is based on:

"Select the best attribute that reduces uncertainty the most."

Uncertainty is measured using:

- Entropy (ID3)
- Gini Index (CART)

ENTROPY (Measure of Impurity)

Definition

Entropy measures disorder or randomness in the dataset.

For binary classification:

$$Entropy(S) = -p_{+}\log_{2}(p_{+}) - p_{-}\log_{2}(p_{-})$$

Where:

- p_{+}= probability of Yes
- p_{-}= probability of No

CART (Classification and Regression Tree)

Developed by:Leo Breiman

CART Uses:

- Gini Index (not entropy)
- Binary splits only

- Works for classification + regression
- Supports numerical features naturally

GINI INDEX

$$Gini = 1 - \sum p_i^2$$

For binary:

$$Gini = 1 - (p_+^2 + p_-^2)$$

ID3 Algorithm – Pseudocode

1. Compute entropy of dataset
2. For each attribute:
   - Compute information gain
3. Choose attribute with highest IG
4. Split dataset
5. Repeat recursively for each subset
6. Stop when:
   - All samples same class
   - No attributes left

Problem Statement:

Goal: Predict whether a player will **Play Tennis (Yes/No)** based on weather conditions.

Step 1: Training Dataset

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Step 2: Compute Entropy of Entire Dataset

Total = 14
Yes = 9
No = 5

$$Entropy(S) = -p_{+}\log_2 p_{+} - p_{-}\log_2 p_{-}$$
$$= -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14}$$
$$= 0.940$$

Entropy(S) = 0.94

Step 3: Information Gain Calculation for Each Attribute

1. **Attribute: Outlook**

Outlook = Sunny

Class	Count
Yes	2
No	3

Entropy = 0.971

Outlook = Overcast

Class	Count
Yes	4
No	0

Entropy = 0

Outlook = Rain

Class	Count
Yes	3
No	2

Entropy = 0.971

Weighted Entropy

$$E(Outlook) = \frac{5}{14}(0.971) + \frac{4}{14}(0) + \frac{5}{14}(0.971)$$
$$= 0.693$$

Information Gain

$$IG(Outlook) = 0.940 - 0.693 = 0.247$$

2. Attribute: Temperature

After calculations:

Temperature	Entropy
Hot	1.000
Mild	0.918
Cool	0.811

Weighted Entropy = 0.911

$$IG(Temperature) = 0.940 - 0.911 = 0.029$$

3. **Attribute: Humidity**

Humidity	Yes	No	Entropy
High	3	4	0.985
Normal	6	1	0.591

Weighted Entropy = 0.788

$$IG(Humidity) = 0.940 - 0.788 = 0.152$$

4. **Attribute: Wind**

Wind	Yes	No	Entropy
Weak	6	2	0.811
Strong	3	3	1.000

Weighted Entropy = 0.892

$$IG(Wind) = 0.940 - 0.892 = 0.048$$

Step 4: Compare Information Gains

Attribute	Information Gain
Outlook	**0.247**
Humidity	0.152
Wind	0.048
Temperature	0.029

Highest Gain = Outlook

So, **Outlook becomes the Root Node.**

After selecting Outlook as root, we MUST recursively apply ID3 on each branch.

Step 5: Split on Root Node = Outlook

We now create 3 branches:

- Outlook = Sunny
- Outlook = Overcast
- Outlook = Rain

We solve each branch separately.

1. **Branch 1: Outlook = Overcast**

Day	Play Tennis
D3	Yes
D7	Yes
D12	Yes
D13	Yes

All = Yes

Entropy = 0

This becomes a **Leaf Node → Play = Yes**

No further splitting needed.

2. **Branch 2: Outlook = Sunny**

Subset (5 samples):

Day	Temperature	Humidity	Wind	Play
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

Step 1: Entropy of Sunny Subset

Yes = 2
No = 3

$$Entropy(Sunny) = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5}$$
$$= 0.971$$

Now calculate Information Gain for remaining attributes:

(Remaining: Temperature, Humidity, Wind)

1. **Humidity (Sunny subset)**

Humidity = High

Yes	No
0	3

Entropy = 0

Humidity = Normal

Yes	No
2	0

Entropy = 0

Weighted Entropy

$$E = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$
$$IG = 0.971 - 0 = 0.971$$

HUGE gain!

2. **Temperature (Sunny subset)**

Hot → 2 No → Entropy = 0
Mild → 1 Yes, 1 No → Entropy = 1
Cool → 1 Yes → Entropy = 0

Weighted Entropy:

$$E = \frac{2}{5}(0) + \frac{2}{5}(1) + \frac{1}{5}(0)$$
$$= 0.4$$
$$IG = 0.971 - 0.4 = 0.571$$

3. **Wind (Sunny subset)**

Weak → 1 Yes, 2 No → Entropy = 0.918
Strong → 1 Yes, 1 No → Entropy = 1

Weighted:

$$E = \frac{3}{5}(0.918) + \frac{2}{5}(1)$$
$$= 0.951$$
$$IG = 0.971 - 0.951 = 0.020$$

Compare Gains (Sunny Subset)

Attribute	Information Gain
Humidity	**0.971**
Temperature	0.571
Wind	0.020

Best split = **Humidity**

Sunny Branch becomes:

If Humidity = High → No
If Humidity = Normal → Yes

No further splitting required (pure nodes).

Branch 3: Outlook = Rain

Subset (5 samples):

Day	**Temperature**	**Humidity**	**Wind**	**Play**
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

Yes = 3
No = 2

$$Entropy(Rain) = 0.971$$

Calculate IG for remaining attributes:

(Remaining: Temperature, Humidity, Wind)

1. **Wind (Rain subset)**

Weak:

Yes	**No**
3	0

Entropy = 0

Strong:

Yes	**No**
0	2

Entropy = 0

Weighted:

$$E = \frac{3}{5}(0) + \frac{2}{5}(0) = 0$$
$$IG = 0.971 - 0 = 0.971$$

Perfect split!

2. **Temperature (Rain subset)**

Mild → 2 Yes, 1 No → Entropy = 0.918
Cool → 1 Yes, 1 No → Entropy = 1

Weighted:

$$E = \frac{3}{5}(0.918) + \frac{2}{5}(1)$$
$$= 0.951$$
$$IG = 0.971 - 0.951 = 0.020$$

3. **Humidity (Rain subset)**

High → 1 Yes, 1 No → Entropy = 1
Normal → 2 Yes, 1 No → Entropy = 0.918

Weighted:

$$E = 0.951$$
$$IG = 0.020$$

Compare Gains (Rain Subset)

Attribute	Information Gain
Wind	**0.971**
Temperature	0.020
Humidity	0.020

Best split = **Wind**

Step 6: Final Decision Tree

```
                 Outlook
            /       |        \
         Sunny   Overcast    Rain
           |        |         |
       Humidity    Yes       Wind
        /    \               /    \
     High   Normal        Weak   Strong
      No     Yes           Yes     No
```

Final Decision Rules

1. If Outlook = Overcast → Play = Yes
2. If Outlook = Sunny AND Humidity = High → No

3. If Outlook = Sunny AND Humidity = Normal → Yes
4. If Outlook = Rain AND Wind = Weak → Yes
5. If Outlook = Rain AND Wind = Strong → No